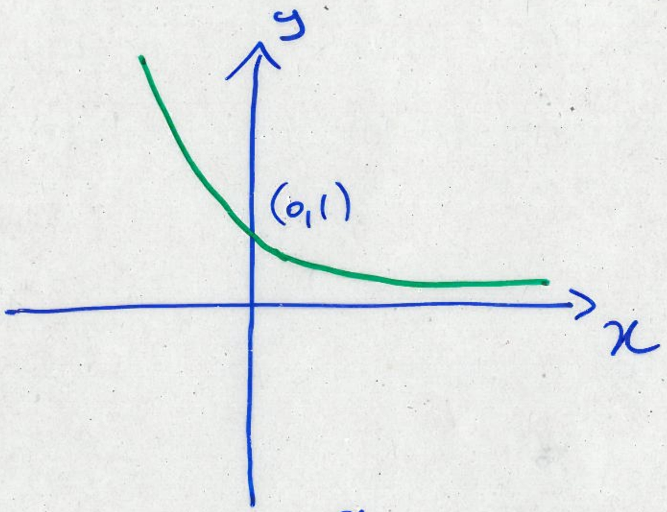
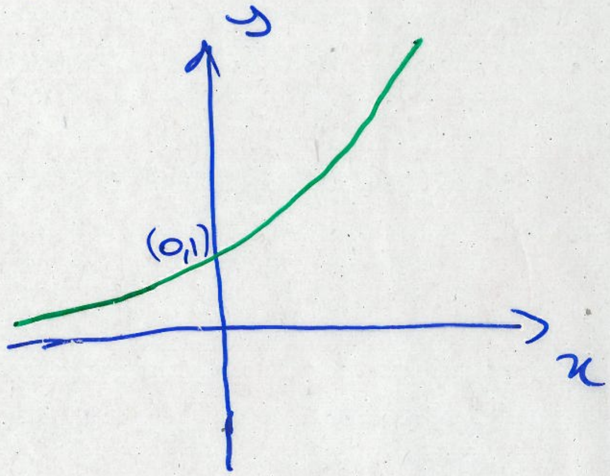
1.7 Exponential functions

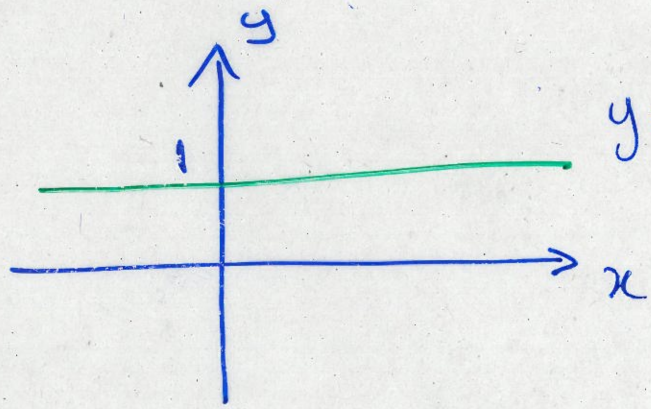
$$f(x) = a^x \quad , \quad a > 0 \quad , \quad a \in \mathbb{R}$$



(a) $y = a^x$, $0 < a < 1$



(b) $y = a^x$, $a > 1$



$y = 1^x$

$R_f = (0, +\infty)$, if $a \neq 1$

$D_f = \mathbb{R}$

<u>Law of exponents</u>: If a and b are positive numbers and x and y are any real numbers, then

$$a^{x+y} = a^x \cdot a^y \qquad (a^x)^y = a^{xy}$$

$$a^{x-y} = \frac{a^x}{a^y} \qquad (ab)^x = a^x \cdot b^x$$

Example 1 : Rewrite and simplify

(a) $\dfrac{3^{-4}}{4^{-8}}$; (b) $\dfrac{1}{\sqrt[5]{x^3}}$; (c) $b^8(2b)^3$

(d) $\dfrac{\sqrt{a\sqrt{b}}}{\sqrt[4]{ab}}$; (e) $\dfrac{3^x(3^x)^6}{3^{4x}}$; (f) $\dfrac{(e^x e^y)^4}{(e^e)^5}$

Example 2 : Sketch on the same graph:

$y_1 = 4^x$, $y_2 = 2^x$ and $y = 10^x$

Example 1 :

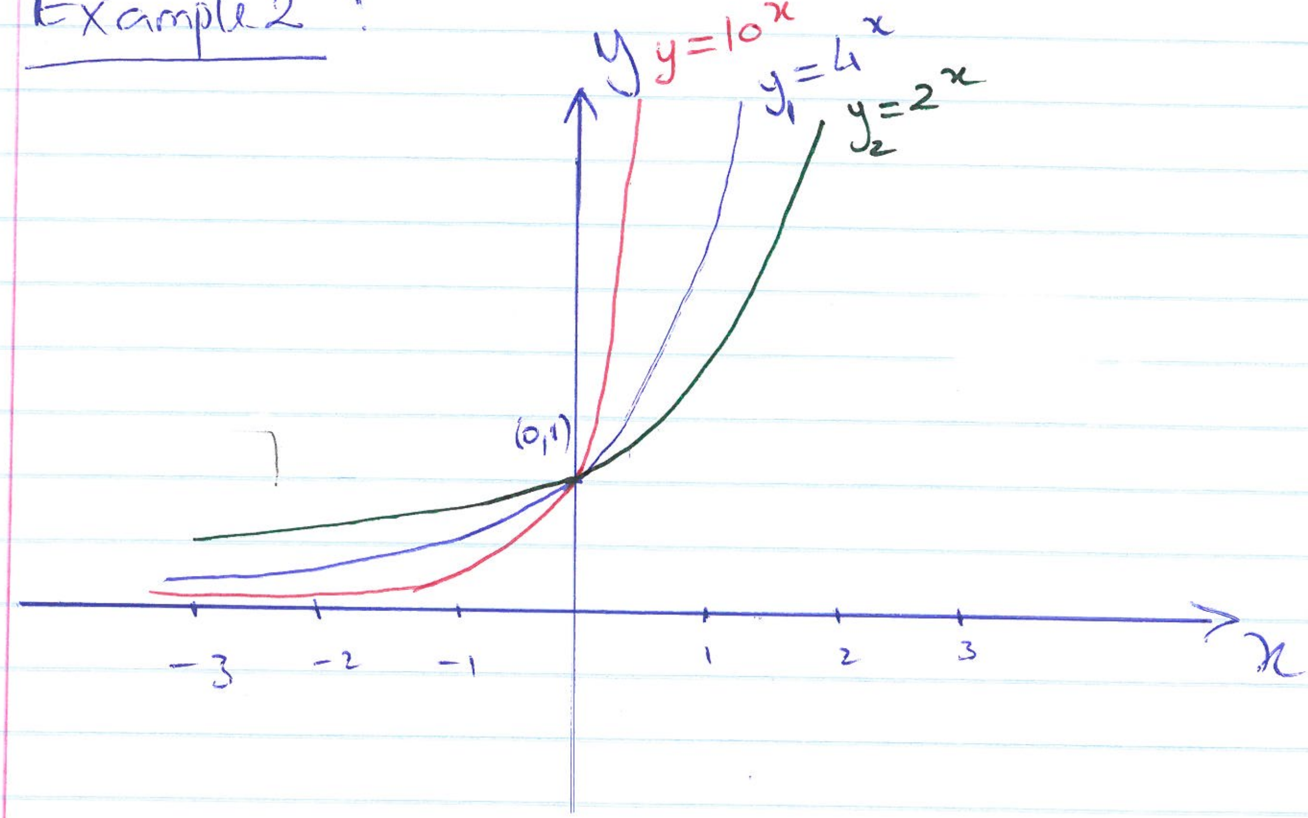
(a) $\frac{3^{-4}}{4^{-8}} = \frac{4^8}{3^4}$

(b) $\frac{1}{\sqrt[5]{x^3}} = \frac{1}{x^{\frac{3}{5}}} = x^{-3/5}$

(c) $b^8 (2b)^3 = b^8 \times 8 . b^3 = 8 b^{11}$

(d) $\frac{\sqrt{a\sqrt{b}}}{\sqrt[4]{ab}} = \frac{(ab^{1/2})^{1/2}}{(ab)^{1/4}} = \frac{a^{1/2} b^{1/4}}{a^{1/4} b^{1/4}} = a^{\frac{1}{2}-\frac{1}{4}} = a^{\frac{1}{4}}$

Example 2 :



$y_1 = 4^x > y_2 = 2^x$ for all $x > 0$

$y_1 = 4^x < y_2 = 2^x$ for all $x < 0$

Hence If $a > b > 0$

$$\boxed{\begin{array}{l} a^x \geq b^x \text{ if } x \in [0, +\infty) \\ a^x \leq b^x \text{ if } x \in (-\infty, 0] \end{array}}$$

Example 1 (continues)

(e) $\dfrac{3^x \cdot (3^x)^6}{3^{4x}} = 3^x \cdot \cancel{3}^{6x} \cdot 3^{-4x} = 3^{7x-4x} = 3^{3x} = 27^x$

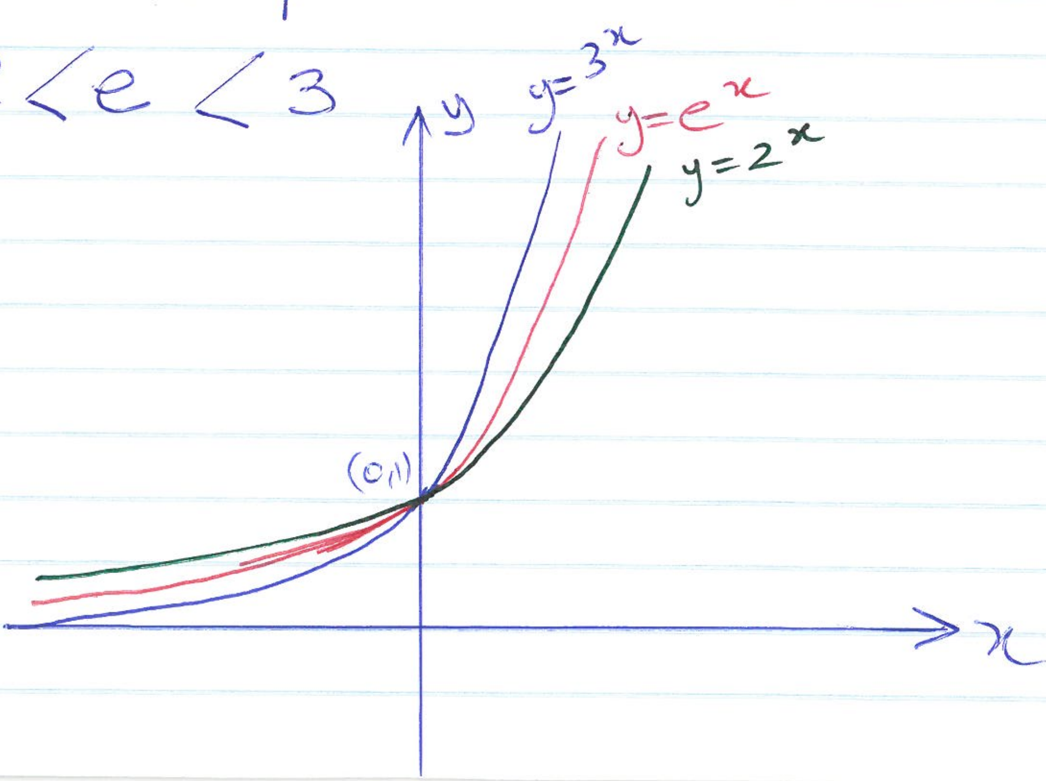
(f) $\dfrac{(e^x \cdot e^y)^4}{(e^e)^5} = \dfrac{e^{4(x+y)}}{e^{5e}} = e^{4(x+y)-5e}$

The number "e" in example 1 (f) is a special irrational number.

$e \approx 2{,}71828$

Therefore $f(x) = e^x$ is called the natural exponential function.

$2 < e < 3$

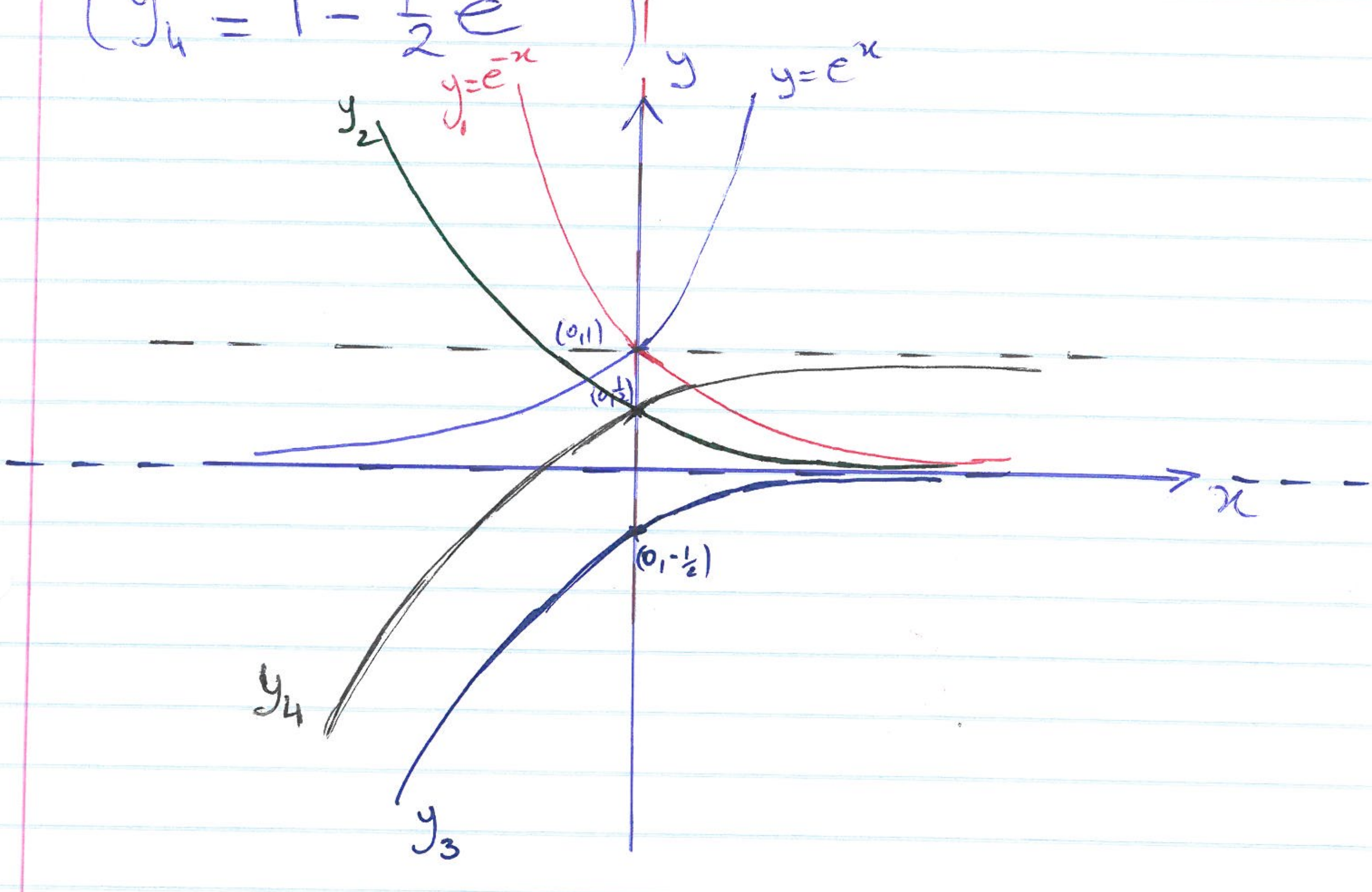


<u>Example 3</u> : Using the graph of $y = e^{x}$ sketch $y = 1 - \frac{1}{2}e^{-x}$.

∴ <u>steps</u>: $y_1 = f(-x)$; $y_2 = \frac{1}{2}f(-x)$; $y_3 = -\frac{1}{2}f(-x)$

$y_4 = 1 - \frac{1}{2}f(-x)$

$$\left(y_4 = 1 - \tfrac{1}{2}e^{-x} \right)$$



Example 4 : sketch $y = e^{(x-3)}$